Gaussian Splatting and Neural Rendering for Dynamic 3D Scene Reconstruction

Abstract

The objective of image-based three-dimensional (3D) scene reconstruction is to convert a collection of two-dimensional (2D) photographic views into a robust, digital 3D model capable of being computationally processed, analyzed, and manipulated from novel viewpoints. Historically, the computer vision and graphics communities relied on explicit representations, such as dense point clouds and polygonal meshes generated via Structure-from-Motion (SfM) and Multi-View Stereo (MVS). However, these classical techniques frequently struggled with view-dependent optical phenomena and texture-less regions, failing to achieve true photorealism. The advent of Neural Radiance Fields (NeRF) initiated a profound paradigm shift by modeling scenes as continuous volumetric functions using deep multilayer perceptrons (MLPs), enabling unprecedented novel-view synthesis. Nevertheless, extending implicit neural representations to dynamic, time-varying environments introduced prohibitive computational overheads due to the necessity of dense ray-marching through four-dimensional (4D) space-time.

Recently, 3D Gaussian Splatting (3DGS) has emerged as a transformative technique that models scenes using millions of explicit, learnable 3D Gaussian primitives paired with a highly optimized, differentiable tile-based rasterizer. This explicit formulation preserves the continuous optimization benefits of volumetric rendering while entirely circumventing the computational bottlenecks of ray-marching, achieving real-time rendering speeds. Consequently, the extension of 3DGS to dynamic scenes—commonly referred to as 4D Gaussian Splatting—represents the current frontier of neural rendering. This comprehensive survey provides an exhaustive, macro-level review of the rapid evolution from static 3DGS to dynamic scene reconstruction. The manuscript systematically categorizes the theoretical underpinnings and landmark developments into distinct structural taxonomies: deformation-based dynamics, native 4D spatio-temporal representations, trajectory and spline-based continuous models, and physics-integrated simulations. Furthermore, it extensively examines domain-specific adaptations, including digital human avatars and autonomous driving environments, alongside critical system-level advancements in compression, streaming, visual SLAM (Simultaneous Localization and Mapping), and robustness against optical degradations such as motion blur. By synthesizing these diverse research trajectories, this report identifies prevailing challenges and maps the future prospects of explicit radiance fields, providing a definitive resource for researchers advancing the boundaries of immersive media, embodied intelligence, and spatial computing.

Introduction

Capturing, reconstructing, and rendering the complex, time-varying dynamics of the physical world from 2D images is a long-standing, fundamental problem in computer vision, robotics, and computer graphics[1]. For decades, the dominant approaches relied on multi-view geometry. Pipelines consisting of Structure-from-Motion (SfM) and Multi-View Stereo (MVS) extracted discrete features across images to triangulate 3D coordinates, ultimately generating explicit geometric models such as dense point clouds or Poisson surface meshes[3]. While geometrically sound, these classical methods frequently failed in regions lacking distinct textures, struggled to model view-dependent effects (such as specular highlights or refractions), and often yielded incomplete topologies that disrupted the photorealism required for novel-view synthesis.

The introduction of Neural Radiance Fields (NeRF) fundamentally disrupted this trajectory by redefining scene representation[5]. Instead of storing discrete geometric primitives, NeRF leverages deep neural networks to parameterize a scene as a continuous, implicit 5D plenoptic function. This function maps 3D spatial coordinates and 2D viewing directions to a volume density and emitted radiance. Through volumetric ray-marching, NeRF achieved unprecedented photorealistic novel-view synthesis[6]. However, the implicit nature of NeRF introduced severe limitations. Generating a single pixel required evaluating a massive Multilayer Perceptron (MLP) hundreds of times along a cast ray, resulting in training times measured in days and inference speeds measured in seconds per frame[9].

Extending these implicit neural representations to dynamic, time-varying scenes exacerbated these computational bottlenecks exponentially[1]. Dynamic scenes require the introduction of a temporal dimension, effectively demanding the reconstruction of a 6D function. Early dynamic neural rendering architectures approached this complexity by utilizing canonical coordinate spaces mapped through complex, non-linear temporal deformation fields[11]. Despite structural innovations like factorized spatial-temporal grids (e.g., K-Planes, HexPlane) that replaced deep MLPs with interpolative feature arrays to accelerate convergence, modeling complex topologies, non-rigid human motions, and highly dynamic autonomous driving scenes using purely implicit volumetric fields remained computationally bounded and structurally opaque[13].

It is within this constrained context that 3D Gaussian Splatting (3DGS) emerged as a paradigm-shifting alternative[16]. By modeling the scene with millions of unstructured, anisotropic 3D Gaussian primitives, 3DGS shifted the rendering paradigm back toward explicit geometric representations without sacrificing differentiability. Paired with a heavily optimized tile-based rasterizer, 3DGS completely bypasses volumetric ray-marching. It projects 3D Gaussians directly onto the 2D image plane, enabling real-time frame rates (often exceeding 100 to 200 frames per second) while maintaining, and frequently surpassing, the high-fidelity synthesis capabilities characteristic of continuous radiance fields[16].

The transition from static 3DGS to dynamic scene reconstruction represents the bleeding edge of computer vision research[2]. Unlike static environments where geometry is rigidly fixed, dynamic 3DGS architectures must overcome severe under-constraint issues: tracking point correspondences over time, handling topological discontinuities as objects fracture or merge, managing massive memory footprints caused by temporal primitive duplication, and modeling physically plausible motions from sparse, monocular, or uncalibrated video inputs[10]. This comprehensive report delivers a systematic synthesis of the dynamic 3DGS literature. By categorizing current methodologies based on their architectural paradigms, mathematical formulations, and domain-specific implementations, this manuscript establishes a unified structural framework for understanding the mechanics, applications, and future trajectories of dynamic neural rendering.

Methods: Theoretical Foundations and Architectural Paradigms

To rigorously comprehend the evolution of dynamic scene reconstruction, it is necessary to first examine the foundational mathematics of static 3DGS, trace the legacy of dynamic implicit fields that inspired contemporary architectures, and classify modern dynamic 3DGS methods into coherent families based on their temporal parameterization strategies.

Foundational Principles of Neural Scene Representation

Before the rapid adoption of explicit primitives, Neural Radiance Fields formalized the mathematics of differentiable volumetric rendering[3]. In NeRF, a camera ray $r(t) = o + td$ is cast from the optical center o along direction d. The expected color $C(r)$ of the pixel is computed by integrating the radiance c weighted by the accumulated transmittance $T(t)$ and volume density $\sigma(t)$ along the ray boundaries $[t_n, t_f]$:

$$C(r) = \int_{t_n}^{t_f} T(t)\sigma(r(t))c(r(t), d)dt$$

where $T(t) = \exp\left(-\int_{t_n}^{t} \sigma(r(s))ds\right)$. Because this integral has no closed-form solution for complex scenes, it is numerically approximated using stratified sampling, querying an MLP at hundreds of discrete points per ray. While variants like Mip-NeRF utilized conical frustums to resolve aliasing[6], and architectures like Instant-NGP, Plenoxels, and TensoRF introduced multiresolution hash grids and tensor decompositions to dramatically accelerate training, the fundamental computational burden of marching through empty space persisted[9]. 3D Gaussian Splatting resolved this by marrying the differentiability of NeRF with the speed of explicit point-based rendering (an evolution of traditional surface splatting)[16]. The 3D geometry of a scene is instantiated as a collection of 3D Gaussian distributions. Each Gaussian is defined by a full 3D covariance matrix Σ and a mean (spatial center) μ:

$$G(x) = \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$$

To maintain mathematical validity during gradient descent (ensuring the covariance matrix remains positive semi-definite), Σ is elegantly factorized into a scale matrix S and a rotation matrix R, parameterized by a 3D scale vector and a 4D unit quaternion respectively, such that $\Sigma = RSS^TR^T$ [16]. The view-dependent appearance of each Gaussian is parameterized by

Spherical Harmonics (SH) coefficients, while an opacity scalar α dictates its density. During the forward rendering pass, a viewing transformation W and the Jacobian of the affine approximation of the projective transformation J are applied to project the 3D covariance into a 2D image-space covariance Σ':

$$\Sigma' = JW\Sigma W^T J^T$$

The rasterizer then sorts the projected 2D Gaussians by depth and performs front-to-back alpha blending to compute the final pixel color C:

$$C = \sum_{i \in N} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j)$$

This explicitly localized formulation naturally avoids empty space, allowing for adaptive density control where primitives are selectively cloned, split, or pruned based on positional gradients[2]. Early improvements to static 3DGS, such as Scaffold-GS, utilized neural anchor points to spawn Gaussians dynamically, while SuGaR aligned Gaussians to extracted surfaces for precise mesh reconstruction[2]. However, mapping these discrete primitives across time introduced an entirely new class of mathematical challenges.

Precursors: Implicit Dynamic Neural Radiance Fields

The conceptual frameworks utilized in modern dynamic 3DGS heavily borrow from the precursor era of dynamic implicit fields. The pioneering D-NeRF and Nerfies architectures bypassed the impossible task of training independent NeRFs for every video frame by introducing a canonical-space paradigm[11]. In these models, the scene is permanently anchored in a static "canonical" space. A separate deformation network T takes a spatial coordinate x and a timestamp t from the observation space and maps it back to the canonical space:

$$T : (x, t) \to x'$$

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To handle the profound ambiguities inherent in observing non-rigid deformations from monocular video, Nerfies injected elastic regularizations, penalizing deformations that violated physical rigidity constraints[12]. However, continuous deformation fields fundamentally fail when topological changes occur—such as a person opening their mouth or objects fracturing—because continuous warping cannot tear space. HyperNeRF addressed this by mapping the scene into a higher-dimensional ambient hyperspace, representing dynamic topological changes as varying slices through this elevated dimensionality[24]. Simultaneously, methods like Neural Scene Flow Fields (NSFF) mapped temporal dynamics by explicitly predicting 3D scene flow (velocity vectors) to enforce consistency across adjacent frames[8]. As the community sought to overcome the severe training latencies of dynamic MLPs, spatio-temporal feature grids revolutionized the field. Architectures like K-Planes and HexPlane decomposed the 4D space-time volume into orthogonal 2D feature planes, while TiNeuVox utilized time-aware neural voxels[13]. These grid-based decompositions compressed the temporal dynamics into efficiently queryable structures, dramatically accelerating training. Yet, because they remained bound to volumetric ray-marching, true real-time rendering of dynamic scenes remained elusive until the integration of explicit Gaussian splatting[10].

Feature / Paradigm	**Implicit Dynamic NeRFs**	**Spatio-Temporal Feature Grids**	**Dynamic 3D Gaussian Splatting**
Primary Representation	Multilayer Perceptrons (MLPs)	Orthogonal 2D Planes / Voxels	Explicit Anisotropic 3D Primitives
Rendering Mechanism	Volumetric Ray-Marching	Volumetric Ray-Marching	Differentiable Tile-based Rasterization
Rendering Speed	$\sim$ 0.1 - 1 FPS	5 - 15 FPS	60 - 200+ FPS
Empty Space Handling	Inefficient (Requires sampling)	Moderate (Requires sampling)	Highly Efficient (Inherently avoided)
Editability	Near Impossible	Difficult	High (Explicit positional control)

Deformation-Based Dynamic 3D Gaussian Splatting

The most intuitive and widely adopted adaptation of 3DGS to dynamic scenes directly transposes the canonical-space deformation paradigm of D-NeRF onto explicit Gaussian primitives[10]. In deformation-based 3DGS, a universal set of canonical 3D Gaussians is maintained in memory. A specialized deformation network is then trained to predict temporal offsets for the mean positions, rotations, and scales ($\Delta\mu, \Delta R, \Delta S$) of each primitive at any given timestamp[17].

In frameworks such as Deformable 3D Gaussians, an MLP evaluates these offsets frame-by-frame, effectively warping the entire point cloud to match the observed temporal state[25]. However, evaluating an MLP for millions of Gaussians per frame introduces a significant computational bottleneck that threatens to negate the inherent speed of splatting. To alleviate this, the 4D-GS framework proposed by Wu et al. synergizes 3DGS with the HexPlane architecture. By encoding spatial-temporal features into a multi-resolution HexPlane, the deformation field is drastically compressed into queryable 4D neural voxels, bypassing deep MLPs and sustaining rendering speeds of up to 82 FPS[17].

Further structural optimizations focus on sparse control parameterizations to enforce motion coherence. The SC-GS (Sparse-Controlled Gaussian Splatting) framework isolates motion from appearance by utilizing a sparse set of control points, significantly fewer in number than the Gaussians themselves, to learn compact 6-Degree-of-Freedom (DoF) transformation bases[15]. The dense Gaussians are locally interpolated based on these control points using learned weights. Crucially, SC-GS enforces spatial continuity and local rigidity via As-Rigid-As-Possible (ARAP) regularizers[15]. This decoupling not only accelerates optimization but uniquely enables user-controlled motion editing—allowing a user to manipulate a control point and observe physically plausible deformations of the surrounding scene in real-time.

Addressing the structural limitations of a single, universal deformation field, the MoE-GS (Mixture of Experts for Dynamic Gaussian Splatting) framework introduces a novel ensemble approach[1]. Dynamic scenes often feature highly variable motion frequencies—a slow-moving background contrasts sharply with rapidly oscillating foreground objects. Unlike standard MoE models in large language models that route at the feature level to reduce FLOPs, MoE-GS utilizes a Volume-aware Pixel Router. This router projects volumetric Gaussian-level weights into the 2D pixel space via differentiable weight splatting, adaptively blending heterogeneous deformation experts. This ensures that high-frequency motions and low-frequency static backgrounds are managed by specialized sub-networks, preserving spatial and temporal coherence without spectral conflict[1].

Native 4D and Spatio-Temporal Gaussians

An elegant structural alternative to the canonical deformation paradigm is the native 4D approach, which conceptualizes the dynamic scene holistically as a single spatio-temporal volume rather than a sequence of warped 3D states. In this paradigm, primitives are inherently four-dimensional, defined by a 4D mean $\mu = (\mu_x, \mu_y, \mu_z, \mu_t)$ and a 4×4 space-time covariance matrix Σ_{4D} [10].

To render an image at a specific queried timestamp t, the mathematical projection of the 4D Gaussian involves a slicing operation along the temporal axis. This slicing mathematically yields two distinct components: a conditional 3D Gaussian representing the spatial extent of the primitive at that exact instant, and a marginal 1D Gaussian that modulates its temporal opacity[30]. This formulation is highly advantageous because it inherently models the physical transience of objects. As the queried time t moves further from the Gaussian's temporal mean μ_t, the marginal 1D Gaussian exponentially decays the opacity, allowing primitives to seamlessly fade into or out of existence, organically resolving the topological tearing issues that plague continuous deformation fields[30]. Furthermore, frameworks like Real-time 4DGS pair this with 4D Spherindrical Harmonics to capture temporally evolving specular highlights and view-dependent appearances[30].

A recognized systemic challenge in native 4D methods is the rapid proliferation of short-lifespan primitives. Because each Gaussian only represents a localized window of time, long video sequences cause an explosion in the number of primitives, resulting in unsustainable storage overheads that can exceed several gigabytes for a few seconds of video. The 4DGS-1K framework explicitly targets this temporal redundancy by introducing a Spatial-Temporal Variation Score[32]. This scoring metric systematically identifies and prunes redundant short-span Gaussians, forcing the network to optimize a sparser set of longer-lifespan primitives. By storing active masks across sequential frames to avoid processing inactive Gaussians during rasterization, 4DGS-1K shrinks the required primitive count drastically, achieving frame rates exceeding 1,000 FPS[32].

Expanding spatio-temporal representations into the frequency domain, the EndoWave model directly optimizes 4D primitives for complex, highly non-rigid environments, such as endoscopic surgical scenes[34]. By incorporating multi-resolution rational wavelet supervision alongside optical flow-derived geometric constraints, EndoWave ensures that the highly ambiguous, textureless tissue deformations are reconstructed with strict structural coherence, bypassing the artifacts common to unconstrained 4D optimization[34].

Trajectory, Flow, and Spline-Based Parameterizations

Rather than modeling motion through global deformation networks or multi-dimensional volume slicing, trajectory-based methods parameterize the explicit temporal path of each individual 3D Gaussian. This approach treats Gaussians as persistent physical entities traversing through space.

Luiten et al. introduced Dynamic 3D Gaussians by treating primitives as persistent elements that explicitly track 6-DoF motion across time[31]. By enforcing strict local rigidity constraints and mandating that color and scale attributes remain persistent, the Gaussians are forced to naturally track the physical matter of the scene. This formulation remarkably allows dense, occlusion-aware 6-DoF tracking of all scene elements to emerge organically from the photometric rendering loss, without requiring external optical flow inputs[31].

To ensure motion smoothness and drastically compress the temporal parameter space, recent state-of-the-art works employ mathematical curve fitting. The SplineGS framework models the continuous trajectory of dynamic Gaussians using cubic Hermite splines or Non-Uniform Rational B-Splines (NURBS)[33]. By utilizing a highly compact set of trainable control points per primitive, SplineGS can explicitly compute the exact spatial translation and orientation at any continuous fractional timestamp t. This continuous mathematical parameterization avoids the temporal jitter inherent in frame-by-frame discrete optimization[33]. Furthermore, because the spline intrinsically regularizes the motion, SplineGS achieves robust reconstruction without relying on external camera pose estimators like COLMAP for monocular videos, instead executing a joint optimization of both the camera trajectory and the spline control points directly from the photometric and geometric consistencies[33].

Taking continuous trajectory modeling to its logical extreme, the GrowFlow architecture integrates Neural Ordinary Differential Equations (ODEs) directly into the Gaussian parameters[39]. Designed specifically for slow, continuous biological processes such as plant growth, GrowFlow models dynamics not as discrete spatial shifts, but as a continuous vector flow field representing the time-varying derivative over position, scale, and orientation. This allows the framework to seamlessly introduce non-linear geometric expansions over time, capturing structural growth that traditional deformation fields fundamentally cannot support[39].

Physics-Integrated Dynamic Generative Models

The overwhelming majority of neural rendering frameworks focus strictly on photometric accuracy—minimizing the pixel-wise difference between the rendered view and the ground truth. This leaves the underlying physical plausibility of the motion mathematically unconstrained. Recently, a groundbreaking sub-field has emerged that directly fuses Gaussian Splatting with continuum mechanics and physical simulators[41].

The PhysGaussian framework represents a watershed moment in generative simulation by explicitly embedding Newtonian dynamics within the 3D Gaussian kernels[41]. Utilizing the Material Point Method (MPM)—a hybrid Eulerian-Lagrangian technique commonly used in computer graphics physics engines—the very same Gaussian primitives used for visual rasterization act as the discrete Lagrangian particles for mechanical simulation[41]. By evolving the kinematic attributes (such as velocity and strain) alongside the covariance matrices and spherical harmonics using Partial Differential Equations (PDEs), PhysGaussian perfectly achieves the paradigm of "what you see is what you simulate." It generates highly realistic, real-time dynamics for elastic objects, plastic metals, and non-Newtonian fluids without relying on traditional intermediate meshes or surrogate "cage" embeddings[41].

Expanding beyond synthetic generative simulation, frameworks like TRACE aim to learn these physical parameters purely from observational multi-view video, completely devoid of human labels[37]. TRACE formulates each 3D point as a rigid particle and learns a translation-rotation dynamics system, explicitly estimating parameters such as mass and momentum to accurately forecast and extrapolate the future motions of complex, interacting objects[37]. Furthermore, in the domain of embodied artificial intelligence and robotic manipulation, the PhysMani framework leverages a divergence-free Gaussian velocity field optimized online. This architecture embeds the physical laws of the target environment directly into a future-aware action policy model, granting robots real-time intuitive physical reasoning regarding the dynamic objects they are interacting with[38].

Trajectory / Physics Mechanism	Core Mathematical Backbone	Key Advantages	Representative Implementations
Persistent Tracking	6-DoF offsets + Local Rigidity Regularizers	Emergent dense tracking without external optical flow.	Dynamic 3D Gaussians[31]
Continuous	NURBS / Cubic	Extreme temporal	SplineGS[33], T-GS[1]

Splines	Hermite Splines	smoothness; sub-frame interpolation.	
ODE Flow Fields	Neural Ordinary Differential Equations	Models non-linear, continuous structural growth over time.	GrowFlow[39]
Continuum Mechanics	Material Point Method (MPM) / PDEs	Perfect alignment of visual rendering and physics simulation.	PhysGaussian[41], TRACE[37]

Domain-Specific Applications: Digital Human and Face Avatars

The explicit geometric nature of 3DGS provides an ideal substrate for modeling digital humans, successfully bypassing the limitations of dense implicit MLPs in capturing high-frequency, dynamic textures such as clothing wrinkles, facial expressions, and hair[10]. The prevailing methodology marries 3DGS with robust parametric human priors, specifically the SMPL or SMPL-X models, to constrain the massive degree of freedom inherent in articulated human motion.

The HUGS (Human Gaussian Splats) framework initializes its 3D Gaussians directly on the surface of a tracked SMPL body model[43]. To capture non-parametric, user-specific details (e.g., loose clothing and hair that SMPL ignores), the Gaussians are allowed to deviate spatially from the base mesh. A critical challenge in this approach involves maintaining surface integrity and avoiding topological holes during extreme skeletal articulations. HUGS solves this by jointly optimizing Linear Blend Skinning (LBS) weights directly for the Gaussians, allowing the independent primitives to move cohesively and realistically during novel pose synthesis[43].

To achieve even higher fidelity and alleviate the computational complexity of animating millions of independent 3D points in unconstrained space, the Animatable Gaussians framework employs a 2D parameterization strategy[44]. By unrolling a learned human template into front and back canonical 2D Gaussian maps, the framework leverages powerful, pre-trained 2D CNN architectures (like StyleGAN) to predict pose-dependent dynamics and spatial deformations. This projection strategy ensures lifelike dynamic appearance and exceptional generalization to unseen, highly complex poses[44].

Scaling from the body to holistic expressive representations, architectures like ExAvatar and GaussianAvatar integrate facial expressions, detailed hand articulations, and full-body pose into a single differentiable pipeline. ExAvatar utilizes a hybrid surface-mesh formulation where Gaussians act as surface vertices maintaining pre-defined SMPL-X connectivity, explicitly overcoming the severe lack of multi-view 3D observations in short, casual monocular video captures[45]. Finally, to enable interaction with general environments rather than isolated rendering, methods like AHA! (Animating Human Avatars) utilize Gaussian-aligned motion modules and opacity-based collision refinements. This allows seamlessly rendered digital humans to realistically navigate, make physical contact, and interact within heavily occluded, dynamically reconstructed 3D scenes[52].

Domain-Specific Applications: Autonomous Driving and Street Scenes

Modeling dynamic urban streets presents an entirely unique set of geometric challenges: environments are large-scale and unbounded, viewpoints are highly sparse, cameras are outward-facing with minimal overlap, and the scenes are populated by rapidly moving heterogeneous objects[1]. Standard 3DGS, optimized for static, inward-facing object-centric capture, suffers from severe background drift and temporal instability in these scenarios.

The DrivingGaussian framework systematically addresses this by proposing a Composite Gaussian Splatting architecture[50]. The driving environment is hierarchically decomposed: the expansive static background is modeled sequentially using incremental 3D Gaussians as the ego-vehicle navigates the space, while moving vehicles and pedestrians are isolated and reconstructed individually using dynamic Gaussian graphs. Global rendering then recombines these elements, accurately restoring real-world occlusions. To manage the sparse views inherent to autonomous driving sensor rigs, DrivingGaussian directly injects LiDAR priors to initialize the spatial distributions of the Gaussians, recovering highly precise geometric shapes[50].

Recognizing the prohibitive cost and sparsity of LiDAR sensors, the OG-Gaussian framework completely replaces LiDAR initialization with semantic Occupancy Grids (OGs) generated entirely from surround-view cameras via an Occupancy Prediction Network[53]. These grids explicitly separate moving vehicles from the static street infrastructure, driving the dense initialization of the primitives without requiring manual 3D bounding box annotations or expensive depth sensors[53].

For advanced self-supervised modeling that completely avoids object-level trackers, the SplatFlow architecture integrates 3DGS with Neural Motion Flow Fields (NMFF)[56]. By learning continuous temporal motion flows of implicit functions, SplatFlow seamlessly tracks and decomposes static backgrounds and dynamic agents, enforcing strict cross-view consistency over time[56]. Furthermore, Multi-Traversal Gaussian Splatting (MTGS) pushes the boundaries of autonomous view extrapolation by utilizing a multi-traversal dynamic scene graph[3]. MTGS learns a highly accurate shared static geometry from multiple overlapping vehicle trajectories collected over months, while isolating traversal-specific appearance variations (e.g., changes in weather, lighting, or time of day) through color correction nodes governed by learnable spherical harmonics residuals[3].

Compression, Streaming, and Hardware Acceleration

Despite boasting unparalleled rendering speeds, the memory footprint of dynamic 3DGS acts as a severe bottleneck restricting its widespread adoption. Uncompressed 4D scenes can require gigabytes of storage for mere seconds of footage, heavily restricting deployment in AR/VR headsets and web-based streaming architectures[55].

Compression techniques broadly fall into two categories: post-optimization pruning/quantization, and feedforward predictive coding. The TC3DGS (Temporally Compressed 3D Gaussian Splatting) framework introduces a sophisticated temporal relevance-based pruning mechanism. By tracking the opacity and volumetric contribution of primitives across their entire lifespan, TC3DGS safely eliminates redundant, short-lived Gaussians[55]. Concurrently, it employs a gradient-aware mixed-precision quantization strategy, allocating higher bit precision strictly to highly sensitive parameters while aggressively compressing static attributes. Coupled with trajectory simplification algorithms (e.g., Ramer-Douglas-Peucker), TC3DGS achieves up to 67× compression without discernible visual degradation[55]. The OMG4 (Optimized Minimal 4DGS) pipeline further formalizes this with a progressive three-stage architecture: Gaussian Sampling, Gaussian Pruning, and Gaussian Merging, fusing primitives with similar spatio-temporal correlations to yield final model sizes of roughly 3 Megabytes[57].

For highly scalable transmission, standardizing 3DGS to interface with conventional 2D video codecs (like H.264, HEVC, or VVC) is paramount, as hardware decoding for these formats is universally available. The D-FCGS (Feedforward Compression for Dynamic Gaussian Splatting) framework standardizes dynamic scenes into a Group-of-Frames (GoF) structure, relying on traditional I-frames and P-frames. By utilizing sparse control points to extract inter-frame motion tensors, it leverages a dual prior-aware entropy model for highly accurate rate estimation[61]. Alternatively, the PackUV and TV-GS (Triplane Video-based 3DGS) frameworks mathematically project the complex 3D attributes of Gaussians onto sequence-structured 2D multi-layer UV atlases or spatial triplanes[63]. This projection meticulously maps the unstructured 3D point data into a highly correlated 2D video stream, allowing hardware-accelerated 2D codecs to compress the dynamic scene representations at unprecedented ratios, enabling true immersive volumetric video streaming[63].

Robustness to Optical Degradations: Motion Blur and Defocus

Real-world videos, particularly casual footage captured by monocular smartphones or fast-moving robotic cameras, are frequently degraded by motion blur (due to rapid subject or camera movement) and defocus blur (due to depth-of-field effects from large apertures)[46]. Direct optimization of 3DGS on blurred imagery fundamentally destroys the underlying geometry, smearing the primitives into vast, inaccurate shapes to artificially mimic the blurred pixels.

Addressing motion blur, the MSCD-GS (Motion-Separated Cooperative Deblurring) and DyBluRF frameworks intelligently decouple camera-induced background blur from the independent, highly non-linear object motion blur[10]. By modeling the integral average of multiple virtual sharp images across the camera's finite exposure time, MSCD-GS separates the point clouds using dynamic semantic masks. It then deploys a Catmull-Rom spline trajectory to explicitly correct the high-frequency dynamics of the moving objects during the exposure window, effectively reversing the physical blur process[46].

For scenes suffering simultaneously from severe defocus and motion blur, unified architectures explicitly model the physical optical convolutions. These systems introduce blur prediction networks (BP-Nets) that estimate per-pixel blur kernels alongside blur-aware sparsity constraints[17]. By rendering latent cross-time rays and explicitly evaluating the circle of confusion (CoC) for depth-of-field defocus, the 3D Gaussians are forcibly constrained to reconstruct a globally sharp canonical scene. The blur is then re-applied synthetically during the forward pass to match the ground truth, resulting in an underlying 3D representation that demonstrates extraordinary geometric resilience to uncontrolled optical degradations[17].

Challenges and Prospects

While dynamic 3D Gaussian Splatting has decisively resolved the rendering speed limitations that hindered implicit Neural Radiance Fields, several profound challenges remain, dictating the trajectory of future research.

Memory Scaling and Hardware Standards: The combinatorial explosion of primitives required for large-scale and long-duration dynamic environments remains a primary algorithmic hurdle. Although advanced compression techniques (like OMG4, PackUV, and D-FCGS) have demonstrated orders-of-magnitude reductions[60], these compression stages add latency to the encoding phase. The ongoing integration of 3DGS into standardized graphic APIs (e.g., Khronos glTF extensions, OpenUSD) and the development of native hardware-accelerated rasterization cores for 3D Gaussians—similar to dedicated polygon rendering pipelines—are essential for ubiquitous adoption in mobile and wearable edge devices[10].

Extreme Topological Discontinuities: Deformation-based and spline-based 3DGS excel at tracking continuous, non-rigid surfaces but fundamentally struggle with events involving extreme topological changes—such as splashing fluids, fracturing objects, or severe occlusions and disocclusions[10]. While native 4D slicing and temporal opacity masking partially mitigate this, tracking physical persistence and maintaining identity through complete topology breaks remains a largely unsolved mathematical problem[10].

Zero-Shot Generalization and Feedforward Models: The current paradigm heavily relies on costly, per-scene optimization (often taking minutes to hours of gradient descent)[3]. Developing robust feedforward inference models that can ingest a highly sparse set of uncalibrated views and directly output a fully parameterized dynamic 3DGS model in a single forward pass is a critical frontier. Methods leveraging large multi-modal vision foundation models to predict spatial-temporal Gaussian distributions directly will define the next generation of generalizable neural rendering.

Semantic Grounding and Causal Physical Reasoning: Moving beyond mere photometric view synthesis, 3DGS must evolve to support actionable intelligence for robotics and autonomous agents[23]. The integration of Vision-Language Models (VLMs) directly into Gaussian features enables open-vocabulary spatio-temporal querying, allowing autonomous systems to identify and track specific objects via natural language. Furthermore, expanding physics-informed frameworks to encompass causal dynamics (e.g., CausalGS) and multi-object interactions will definitively transition 3DGS from a passive visualization tool into an active, interactive 3D world model[52].

Conclusion

The evolution of dynamic 3D scene reconstruction has undergone a profound paradigm shift over the last half-decade. Neural Radiance Fields initially established the theoretical viability of continuous scene parameterization, proving that neural networks could synthesize photorealistic novel views. However, their reliance on dense implicit queries proved antithetical to the strict latency and scalability requirements of dynamic applications. 3D Gaussian Splatting shattered this bottleneck by returning to an explicit, discrete geometric representation, elegantly leveraging differentiable tile-based rasterization to achieve unprecedented real-time photorealism.

As detailed throughout this survey, the extension of 3DGS into the temporal domain has sparked rapid, multi-faceted innovation. Deformation fields and Native 4D parameterizations have successfully untangled complex motions, while explicit spline and trajectory modeling guarantee smooth temporal coherence. The seamless integration of physics-based continuum mechanics (like the Material Point Method) has effectively blurred the historical boundary between visual rendering and physics simulation. Concurrently, domain-specific adaptations have proven highly effective: parametric LBS models guide high-fidelity digital human avatars, while composite scene graphs and spatial occupancy grids enable the robust modeling of vast, unbounded autonomous driving environments. Crucially, algorithmic compression and video-codec integrations are actively resolving the primary bottleneck of memory consumption, paving the way for standardized free-viewpoint video streaming.

Looking forward, the fusion of explicit Gaussian primitives with large-scale generative priors, multimodal foundation models, and causal physical reasoning will drive the next epoch of 3D vision. Dynamic 3D Gaussian Splatting is no longer merely a novel rendering algorithm; it has matured into a foundational, interoperable substrate for next-generation embodied AI, immersive spatial media, and complex scene understanding.

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